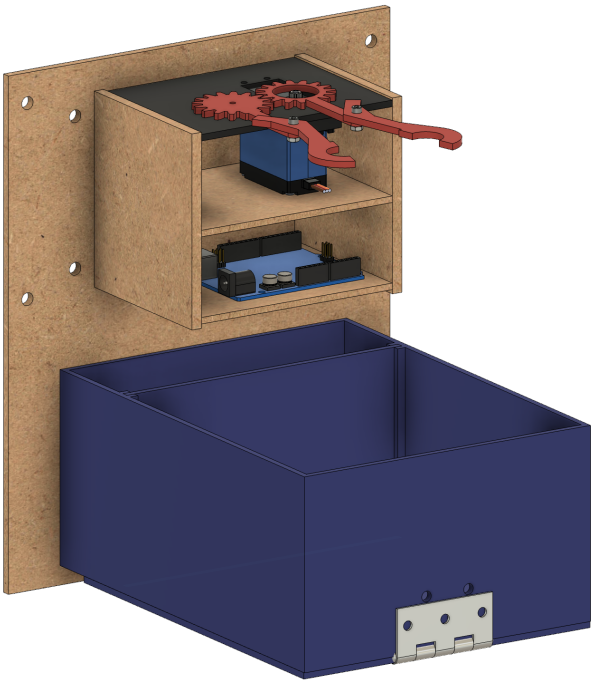
Product Design Project

Final Report



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1 Summary of the approach

1.1 Choice of Mechanism

The gripper mechanism was designed with simplicity and efficiency in mind. It features two half-circle fingers positioned to gently push raspberries down from above. This approach was selected to avoid damaging the berries while ensuring effective handling. Unlike traditional claw-based designs, this mechanism minimizes complexity, focusing on precision and gentle operation. The movement of the gripper is controlled manually via the joystick of the robotic arm to which it is mounted. This ensures that no excessive force is applied, allowing for accurate manipulation of the berries without risking damaging them.

Once collected, the raspberries are deposited in an onboard storage box. The box features a V-shaped opening that allows the berries to slide smoothly into the final collection container. This design choice simplifies the transfer process and minimizes the need for additional handling.

1.2 Choice of Sensors

The gripper integrates a photoresistive sensor to detect raspberry ripeness according to their colour. This choice was made for its ease of integration and fast processing capabilities. Via testing, appropriate thresholds for ripe and unripe berries were established to ensure reliable detection. However, for the photoresistive sensor to function optimally, we integrated a white LED that remains turned ON during operation. The LED provides consistent lighting conditions, which results in the reduction of external light variability, ensuring precise readings from the photoresistor. This setup improved the sensor's accuracy, essential for distinguishing between ripe and unripe berries.

Initially, we considered a camera-based system for the ripeness detection task. However, the slow communication between the camera and the Arduino board via serial as well as the heavy computing required made it impractical, leading us to select the photoresistor solution as a much more efficient alternative.

1.3 Choice of Materials

Our material selection prioritized lightweight, low-cost options to adhere to the project's budget and time constraints. The primary structural components were constructed using Medium-Density Fiberboard (MDF) due to its affordability, availability, and ease of fabrication through laser cutting. For the gripper fingers and other parts with intricate geometries, 3D-printed materials such as PETG were employed. This choice allowed for precise finishes, ensuring a better grip on the berries, which is critical for pushing them down into the box.

Although some aesthetic compromises were made to reduce costs, this combination of MDF and 3D printed components provided a functional and cost-effective solution, which aligns with the project's objectives of practicality and efficiency. In particular, since there are no excessive loads on any parts of our prototype, there was no need for more robust materials.

2 Reflection on the design

The gripper's design, shaped around simplicity, functionality and a search to minimise the number of components, proved to be less reliable than expected. Notably, the photoresistor sensor was highly impacted by the ambient light in operation, even when illuminating the berry with a white LED to attempt reaching constant lightning conditions on the analysed raspberry. In addition, the design of the claw, with the gears a separate component from the claws in order to simplify their design was ultimately a poor choice, as it added an extra point of failure in the claws not always being perfectly aligned.

Nonetheless, despite these weaknesses, we still believe that the overall of the raspberry picking process that we implemented, i.e. pulling downwards from the stem of the raspberry, was strong in its simplicity and in avoiding any damage to the berries.

3 Sustainability and Scalability

Our design is cost-effective, and by reducing the number of components, motors and sensors, we believe it scores well in sustainability analysis. It could however be improved by reducing the size of the 3D-printed plastic collection box, by ensuring a sustainable origin for the wooden MDF parts or by making facilitating recyclability and easy disassembly of the electronic parts.

If our prototype was to be scaled to be used in real-world application, we would modify the design of the compartment of our gripper where the electronical components are installed, ensuring that it is completely closed and waterproof during operation, but also that it can be easily opened for maintenance or repairs. This part of the gripper as well as the fixation board's production may need to be reimagined to avoid hand assembly of the laser-cut parts, for example by using additive production or injection molding.

4 Literature and Patent Research

This patent describes a harvester apparatus and method for separating ripe fruit from harvested plants using a dual sorting system based on colour detection. This approach is designed for large-scale applications like tomato harvesting but lacks the delicacy required for soft fruits like raspberries. In contrast, our system focuses on precision handling and ripeness detection tailored for delicate berries.

Similarly to ours, this patent presents a berry automatic berry picking device with ripeness detection. However, its reliance on centralized control and industrial setup makes it less suitable for portable, small-scale operations. In contrast, our design addresses this gap by emphasizing modularity and adaptability for localized use.

This third patent details a fruit ripeness detection and collection system using weight sensors and a multi-chamber collection apparatus. While innovative, its primary focus is on fruits like oil palm, and it involves methods like filtering by size and weight thresholds. Our device diverges by incorporating photoresistor-based detection and servo-controlled gripping, offering a gentle yet effective solution for raspberry harvesting.